

Mapping the quantum limits of classical simulatability

Leyna Shackleton

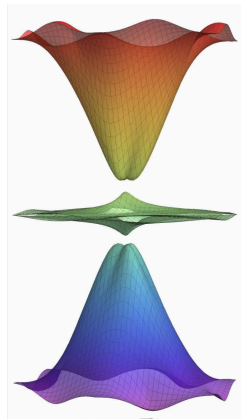
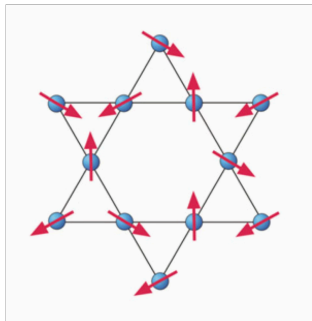
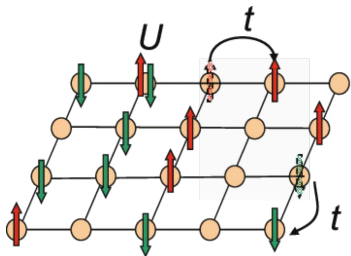
MIT

August 24, 2026

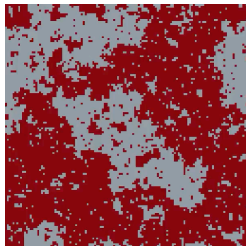
L. Shackleton, *Phys. Rev. Lett.* **136**, 186503 (2026) [arXiv:2509.03708]

L. Shackleton, in progress

Simulating quantum systems is hard



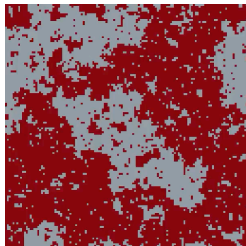
Simulating classical systems is (more) easy



$$Z_{\text{classical}} = \sum_x e^{-E(x)/k_B T}$$

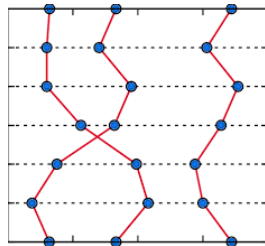
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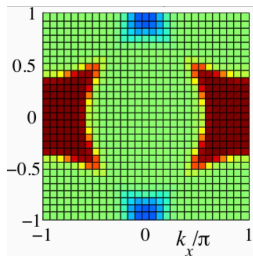
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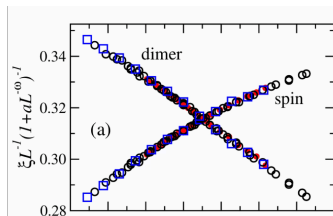
$$Z_{\text{quantum}} = \text{Tr} \left[e^{-\hat{H}/k_B T} \right]$$

Weights can be complex $\Rightarrow$ sign problem

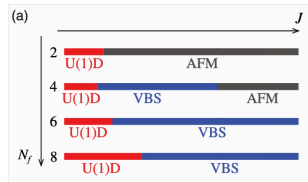
Sign problem-free models contain non-trivial physics



Critical Fermi surfaces
(Berg et al. 2012)



Deconfined criticality
(Sandvik 2007)



Gauge theories
(Xu et al. 2019)

Can a phase have an intrinsic sign problem?

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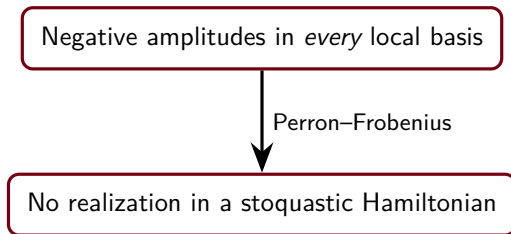
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Perron–Frobenius: a stoquastic $\hat{H}$ has a ground state whose amplitudes are all non-negative in that basis

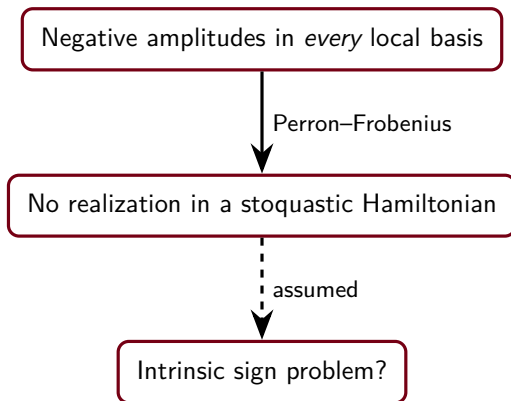
Ground state non-positivity as a proxy

Negative amplitudes in *every* local basis

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No-go theorems for topological phases

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- Most topological orders (Seo 2025)

*“... **398** of **405** suffer from the intrinsic sign problem.”*

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Criterion: **wavefunction non-positivity**

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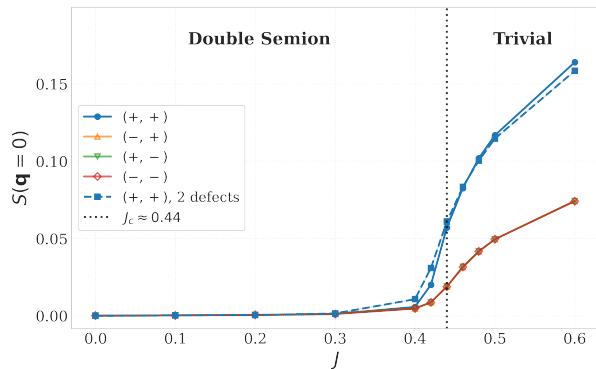
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Double semion $\rightarrow$ trivial transition

Stochastic series expansion and the sign problem

$$Z = \sum_j \sum_n \sum_{\{b_1, \dots, b_n\}} \frac{(-\beta)^n}{n!} \langle z_j | H_{b_n} \cdots H_{b_2} H_{b_1} | z_j \rangle$$

Stochastic series expansion and the sign problem

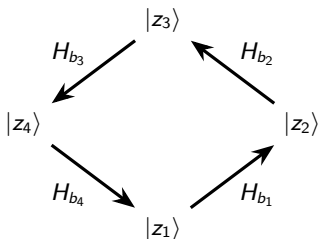
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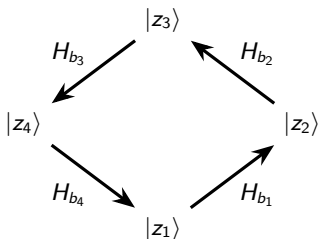
sample *closed* trajectories

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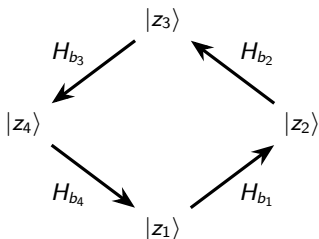
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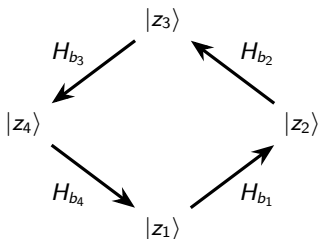


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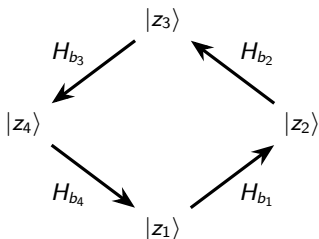
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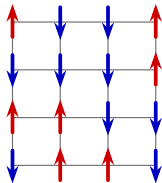


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- Sign problem-free iff stoquastic after phase rotations (Hen 2021)

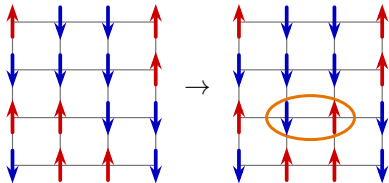
Example: Heisenberg antiferromagnet on bipartite lattices

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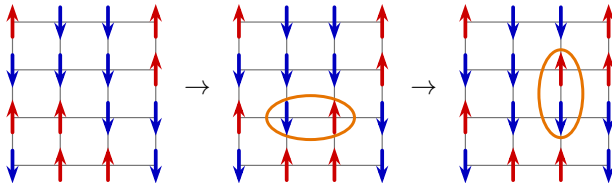
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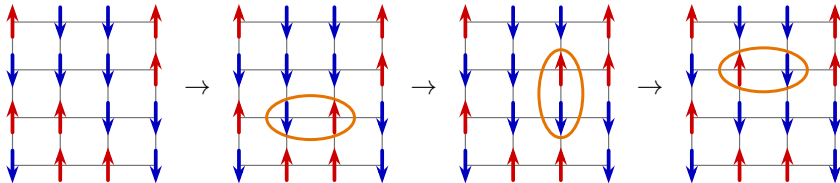
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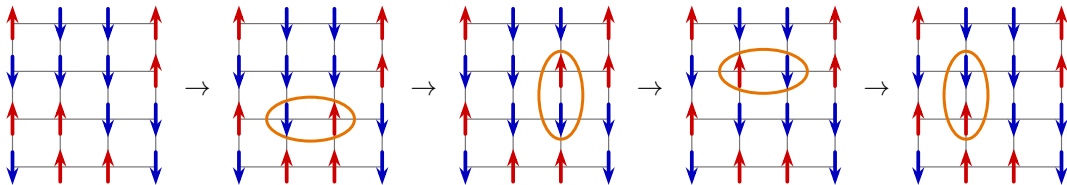
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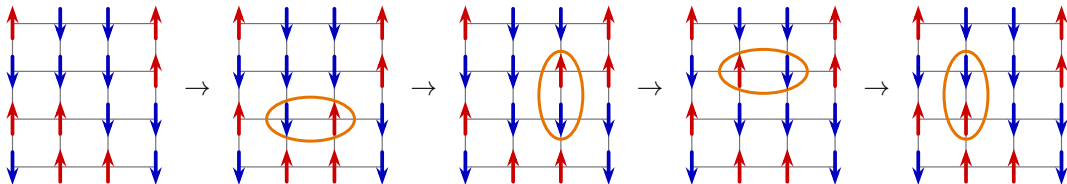
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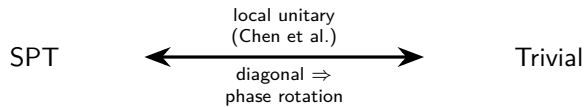
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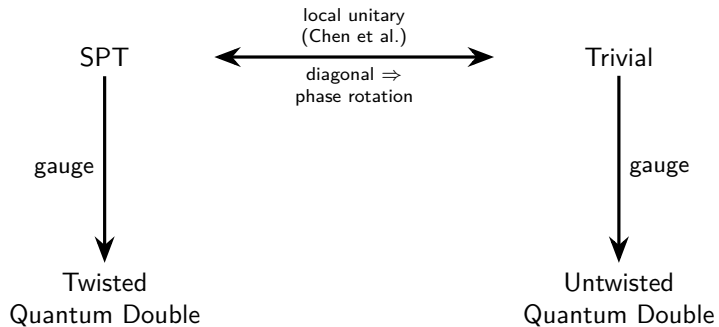


Marshall sign rule: $U = \exp \left[i\pi \sum_{i \in A} \frac{1 - \sigma_i^z}{2} \right]$

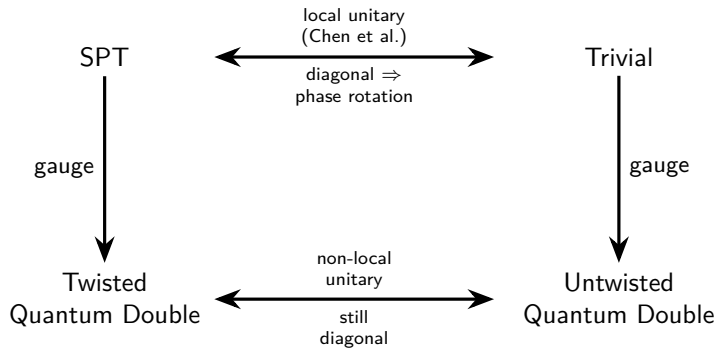
Main idea



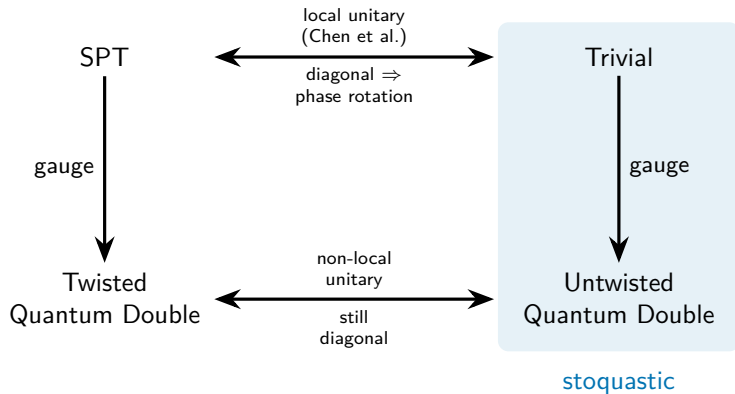
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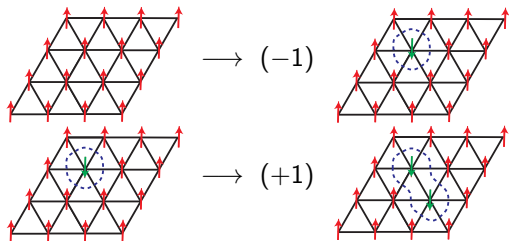


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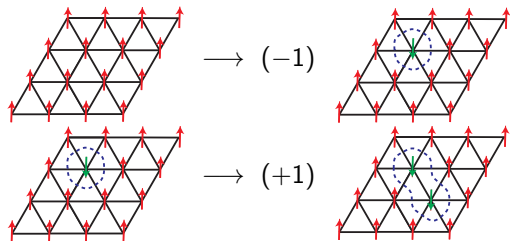
The Levin–Gu model is sign problem-free

$$H_{\text{Levin-Gu}} = - \sum_j \sigma_j^x \prod_{\triangle_{jkl}} i^{\frac{1}{2}} (-1 - \sigma_k^z \sigma_l^z) \equiv - \sum_j A_j$$



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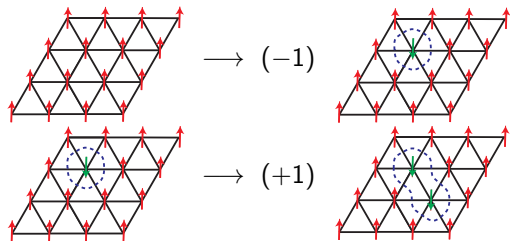
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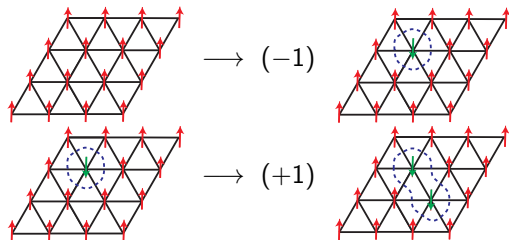
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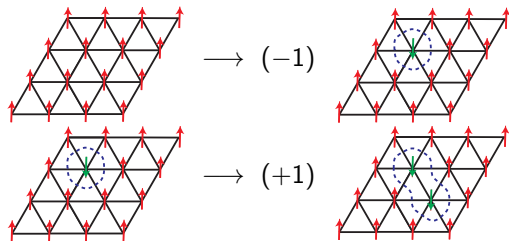
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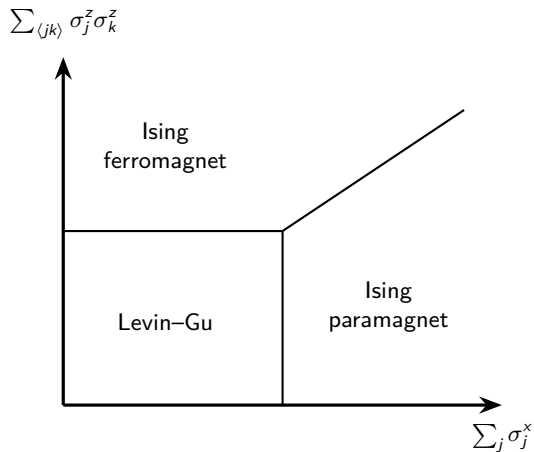
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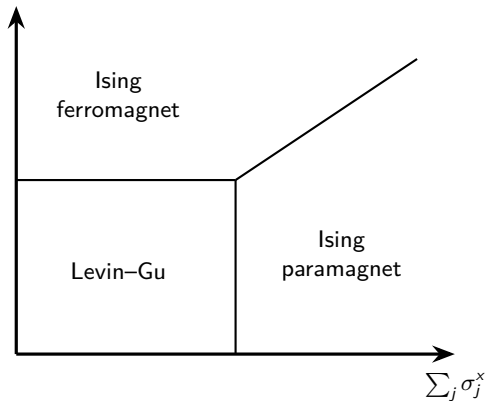
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- *Symmetry-protected* sign problem (Ellison 2021)

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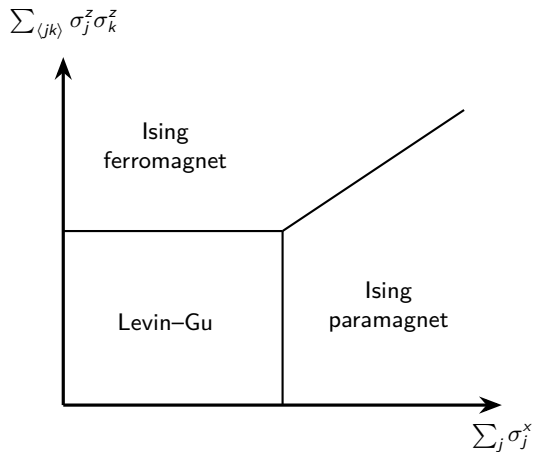
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$$\sum_{\langle jk \rangle} \sigma_j^z \sigma_k^z$$



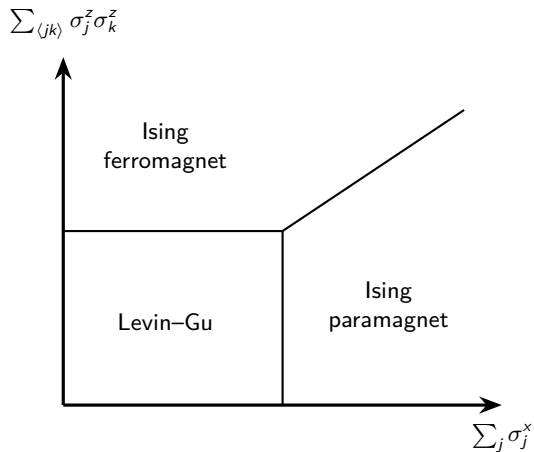
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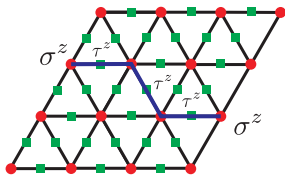
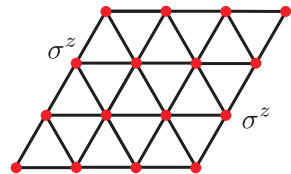
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- $\Rightarrow$ entire phase diagram sign problem-free

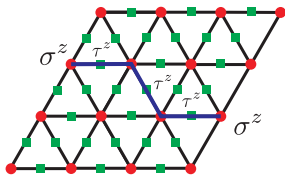
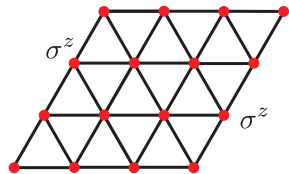
Gauging the Levin–Gu model



$$H_{DS} = -\sum_j \sigma_j^x \prod_{\Delta_{jkl}} i^{\frac{1}{2}} (-1 - \sigma_k^z \tau_{kl}^z \sigma_l^z) - \sum_{\Delta_{jkl}} \tau_{jk}^z \tau_{kl}^z \tau_{lj}^z - \sum_j \sigma_j^x \prod_k \tau_{jk}^x$$

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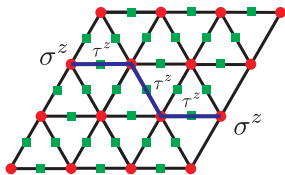
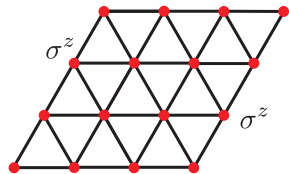


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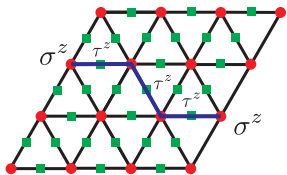
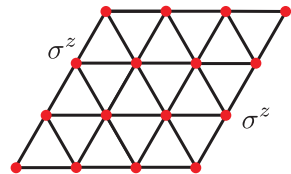


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- $H_{DS} = \tilde{U} H_{TC} \tilde{U}^\dagger$ (requires careful analysis)

Gauging the Levin–Gu model

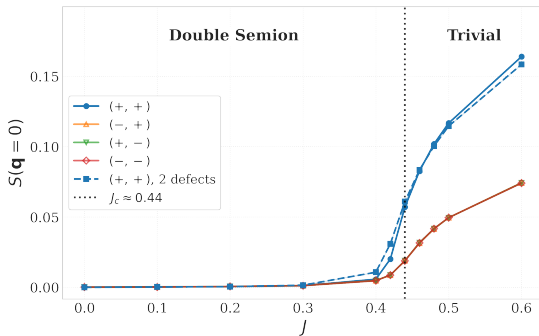
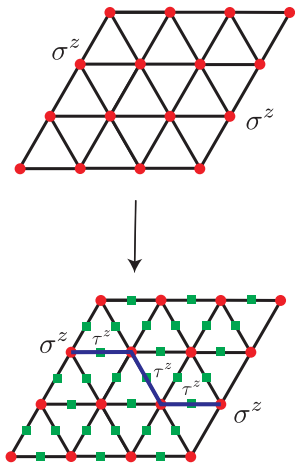


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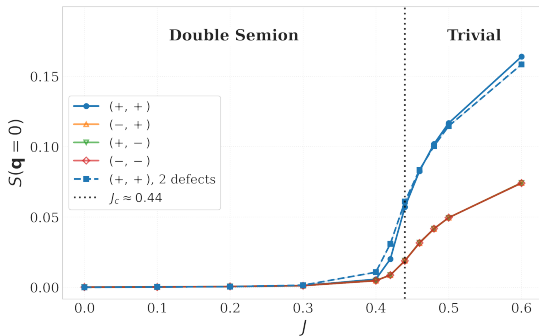
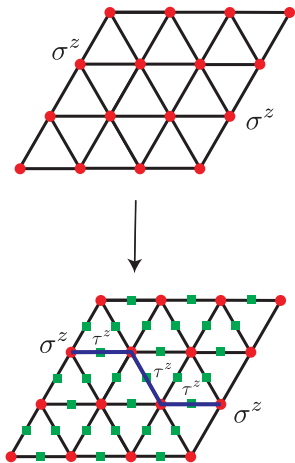
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- U symmetric $\Rightarrow$ gauges to $\tilde{U}$: **diagonal**, non-local
- $H_{DS} = \tilde{U} H_{TC} \tilde{U}^\dagger$ (requires careful analysis)
- $\Rightarrow$ SSE sign structure of H_{TC} and H_{DS} is identical

Gauging the Levin–Gu model

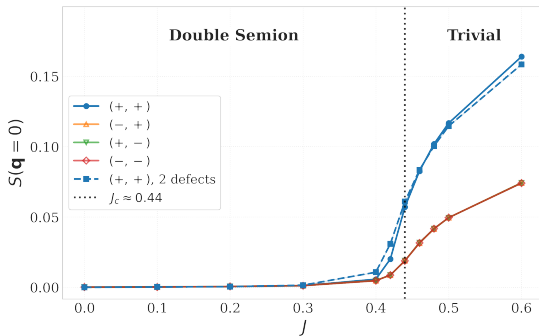
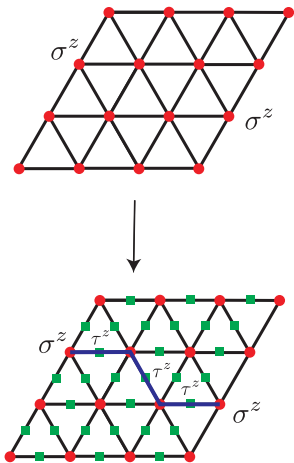


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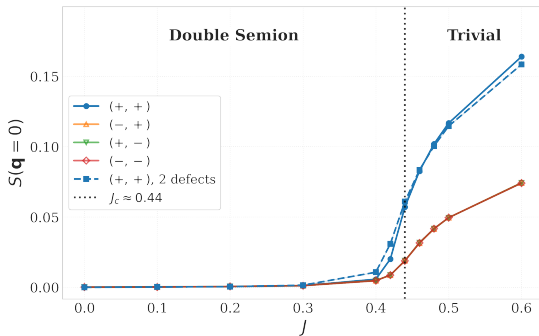
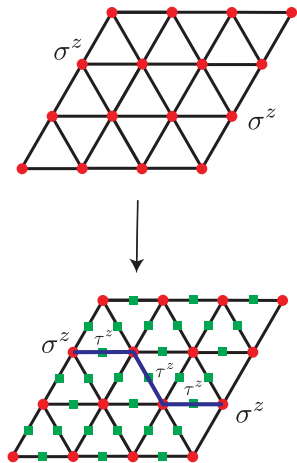
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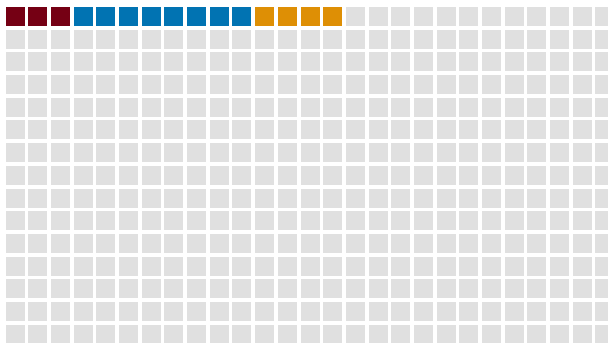
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- Diagonal perturbations are free $\Rightarrow$ condense fluxes, access phase transitions

What are the true limits to simulatability?



■ 3 previously known ■ +8 this work ■ 4 undetermined ■ 390 intrinsic sign problem

How many more?

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What else can be reformulated?

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